

Iris Dataset Exploratory Data Analysis

1. Dataset Overview

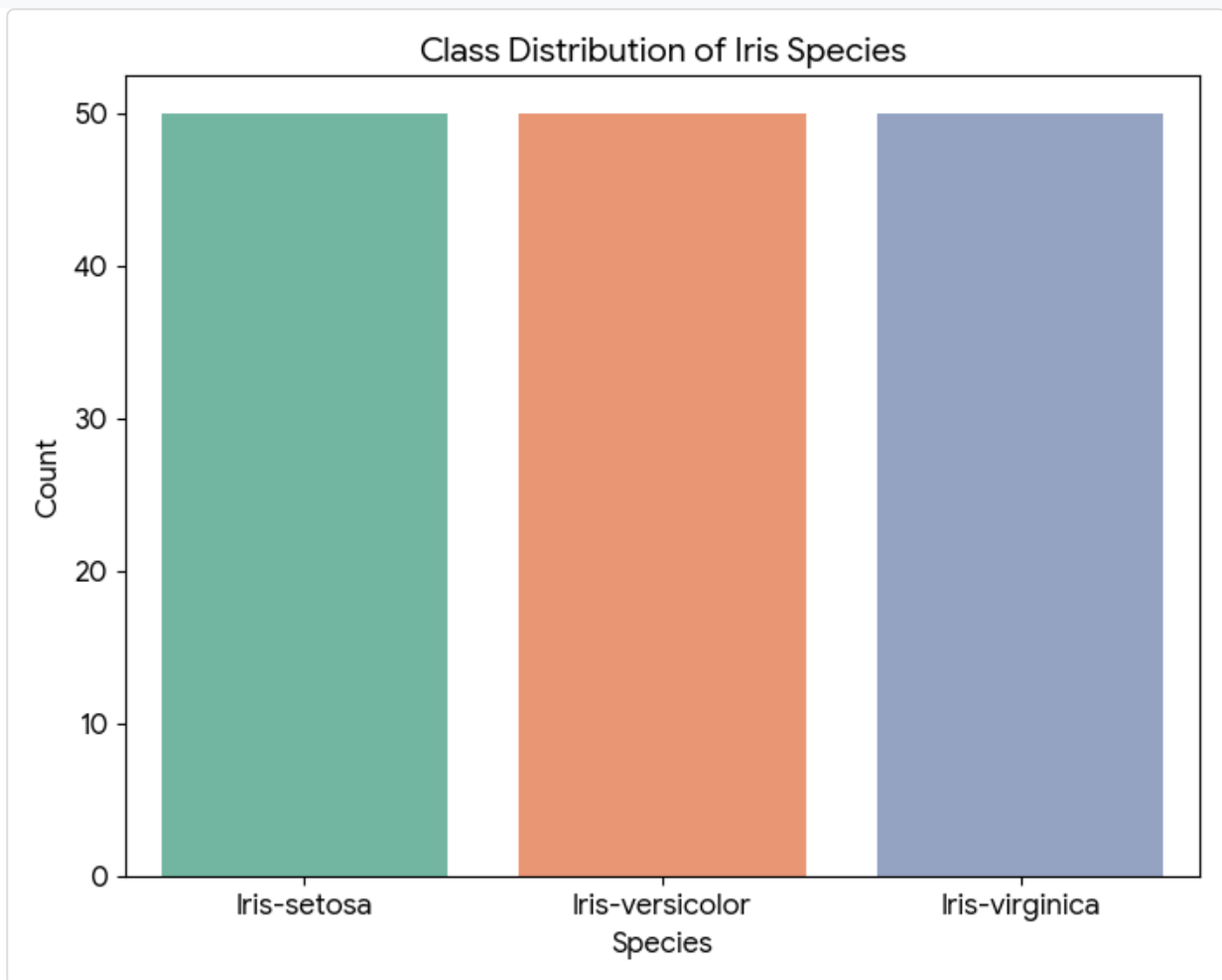
The Iris flower data set is a classic dataset in machine learning and statistics. It contains 150 observations of iris flowers from three different species: Iris-setosa, Iris-versicolor, and Iris-virginica. For each flower, four features were measured: sepal length, sepal width, petal length, and petal width.

Dimension: 150 rows, 5 columns (after dropping the 'Id' column).

Missing Data: The dataset is clean with 0 missing values across all columns.

2. Class Distribution

The dataset is perfectly balanced, with 50 samples for each of the three species.



3. Statistical Summary

Below is the statistical summary of the numerical features in the dataset:

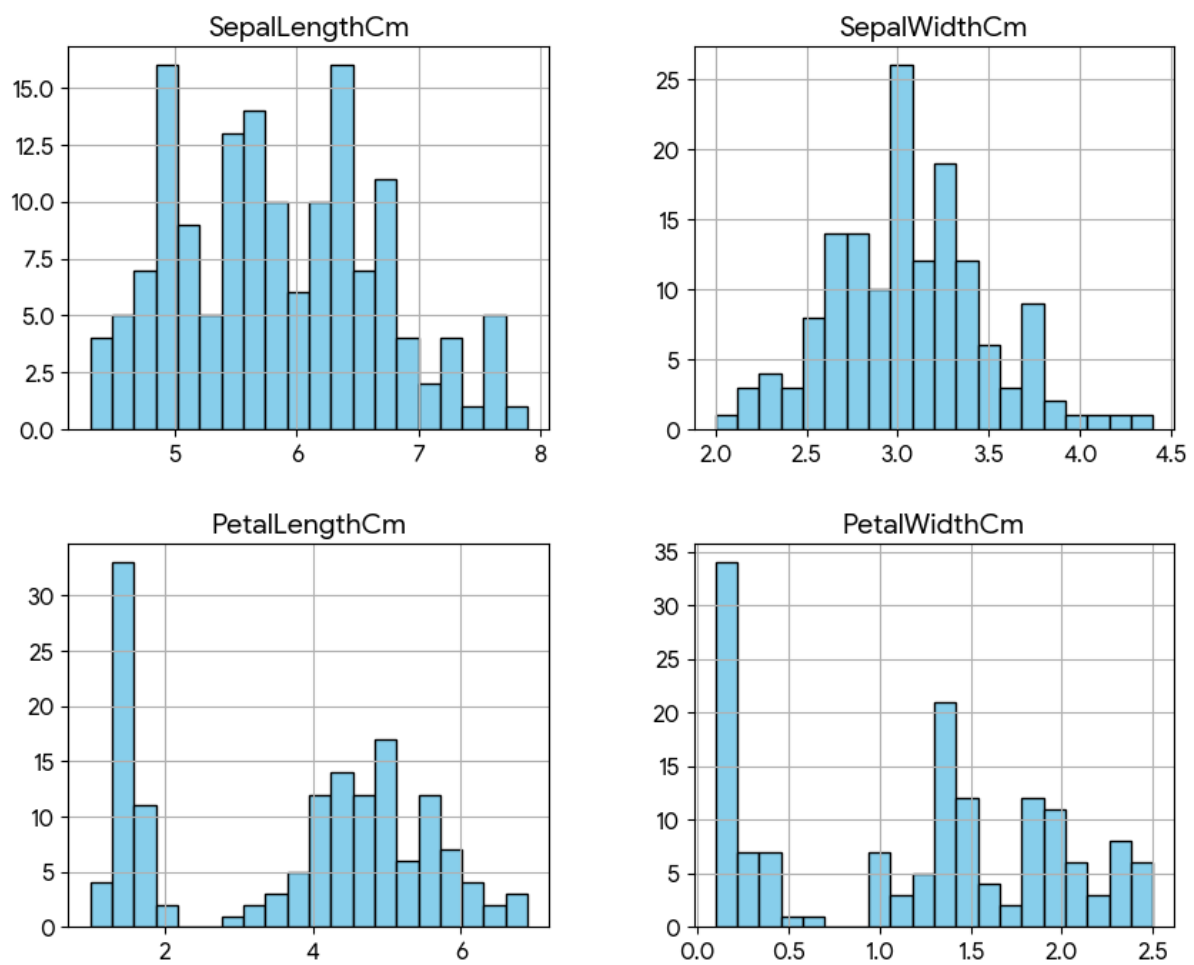
Metric	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.00	150.00	150.00	150.00
mean	5.84	3.05	3.76	1.20
std	0.83	0.43	1.76	0.76
min	4.30	2.00	1.00	0.10
25%	5.10	2.80	1.60	0.30

50%	5.80	3.00	4.35	1.30
75%	6.40	3.30	5.10	1.80
max	7.90	4.40	6.90	2.50

4. Feature Distributions

The histograms below show the distribution of each individual feature across the entire dataset. Notably, Petal Length and Petal Width show bimodal distributions, suggesting these features might be good discriminators for at least one species.

Histograms of Iris Features



5. Pairwise Relationships

The pairplot visualizes the relationships between all pairs of features, colored by species. This reveals clear clustering:

- **Iris-setosa** is easily separable from the other two species, particularly using Petal Length and Petal Width.
- **Iris-versicolor** and **Iris-virginica** have some overlap but are generally distinguishable, again with Petal dimensions being stronger differentiators than Sepal dimensions.

